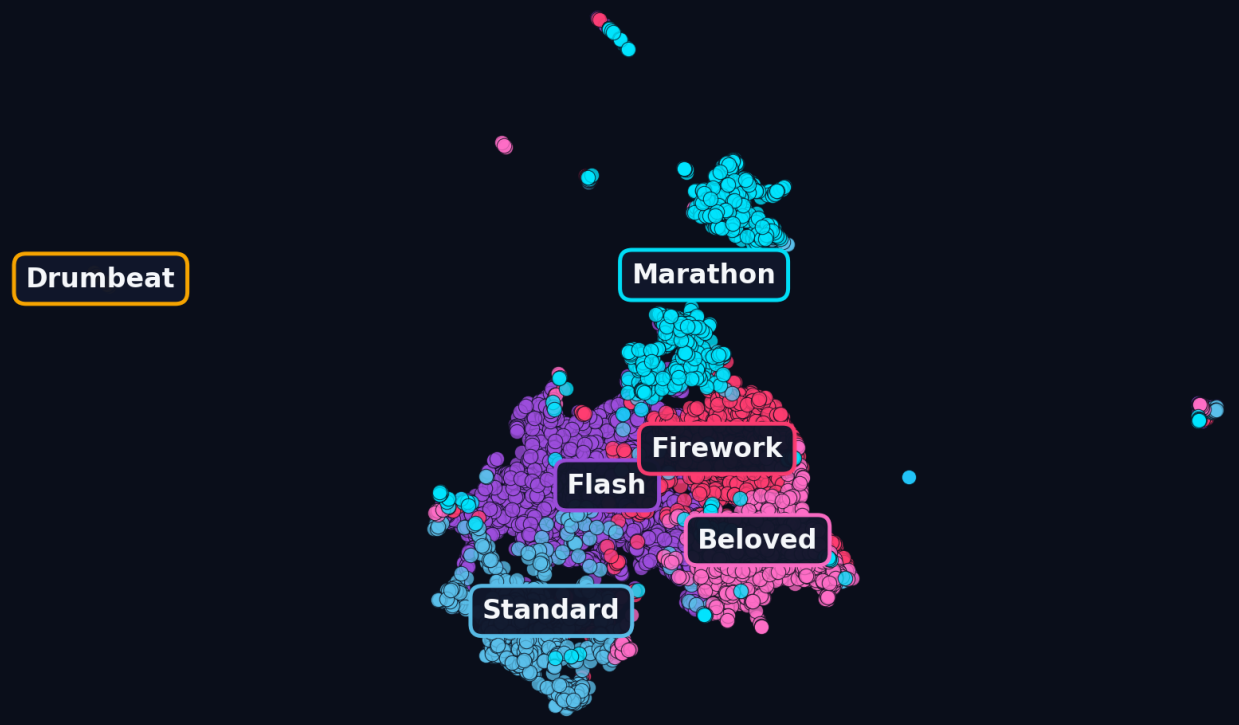
UGDSAI 29 · UNSUPERVISED MACHINE LEARNING · END-TERM PROJECT

What goes viral in India?

A longitudinal study of YouTube Trending — six archetypes, found by machine.



9,100	6	0.55
VIDEOS	ARCHETYPES	METHOD ARI

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Trending is not one phenomenon — it is several

And the difference matters for anyone making, marketing, or studying video.

STICKY

Daily soaps stay on the trending chart for weeks.
Different dynamics entirely.

FLASH

Music drops climb fast, plateau, and slip away in under three days.

LOYAL

Some creators get +200% the like-to-view ratio.
Small, devoted audiences.

OUR QUESTION

Can we discover these archetypes from data alone — and predict which one a video belongs to from the metadata available the moment it first appears on the trending chart?

Three different stakeholders, three different decisions

The same data answers very different questions for each.

CREATOR

A creator deciding whether to invest in long-form content

Marathon and Firework imply opposite production strategies. One rewards depth and consistency; the other rewards timing and reach.

BRAND

A brand planning when to spend ad budget

Firework windows are 24-48 hours. Marathon partnerships pay off over weeks. Knowing the difference avoids burning budget on the wrong creator.

PLATFORM

A platform team studying its recommender

Drumbeat (chart re-entries) and Beloved (engagement spikes) signal different recommender outcomes. Tracking archetype share over time becomes a health metric.

37,352 snapshots, 16,307 unique videos, 7 months

Hybrid: archival India trending data + a fresh live collector validating the pipeline today.

End-to-end pipeline

Every step reproducible from the codebase · runs in 30 seconds locally

Collect (API + archival)
9,100 videos
from 7 months of IN trending

→

Long format store
snapshot rows
video × time

→

Feature engineering
25 features
in 5 themes

→

Scaling + PCA
95% variance
→ 16 dims

→

Clustering (K-Means · k=6)
ARI = 0.55
between methods

→

Profile + name
Marathon, Firework,
Beloved, Drumbeat, ...

Same schema, same code path: live collector for fresh data, archival CSV for depth. Pipeline runs end-to-end in 30 seconds.

37,352	16,307	205	YouTube
SNAPSHOT ROWS	UNIQUE VIDEOS	DAILY SNAPSHOTS	DATA API v3

25 features in 5 themes, all derived from the lifecycle

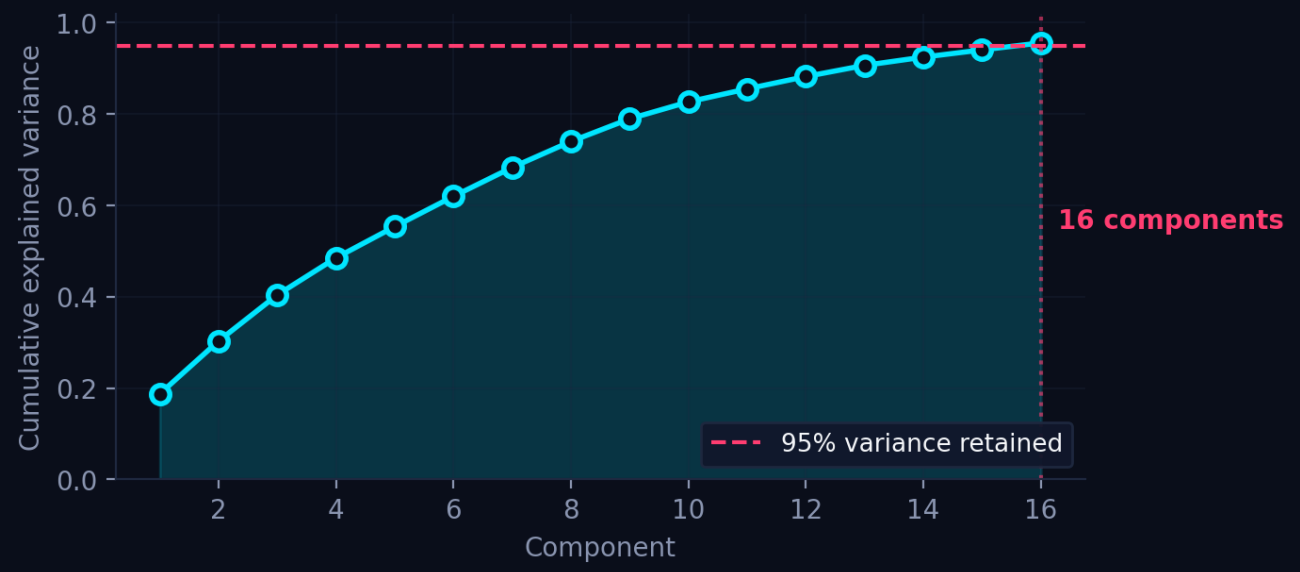
One row per video, not per timestamp — features describe how a video behaved over its life.

VELOCITY	DECAY	RETENTION	ENGAGEMENT	CONTENT
How fast it grew	How fast it died	Sticky vs flash	Audience response	Video / title / channel
`peak_views_per_hour`	`decay_log_slope_48h`	`chart_presence_ratio`	`mean_like_view_ratio`	`duration · is_short`
`mean_views_per_hour`	`half_life_hours`	`rank_volatility`	`mean_comment_view_ratio`	`title length · caps`
`hours_to_first_trend`	`days_observed`	`returned_count`	`comment_like_ratio`	`tag_count · category`
			`engagement_growth`	

Aggregating across each video's snapshots gives features that describe behaviour, not just state.

PCA: 16 components retain 95% of the variance

Engineered features are correlated by design; PCA gives us a clean basis to cluster on.



What the top components capture

PC1 **Velocity ↔ Longevity**

Heavy loadings on peak views/hour and chart presence. The fundamental trade-off.

PC2 **Engagement ↔ Volume**

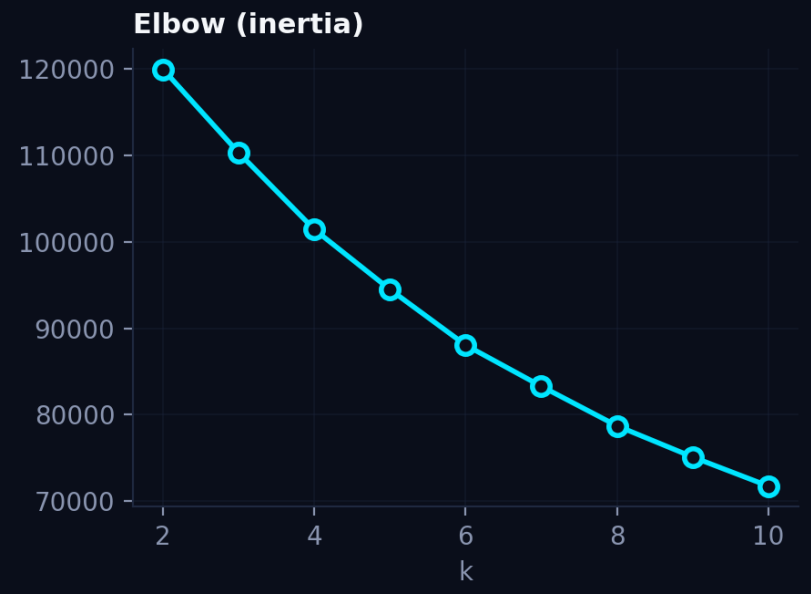
Loadings on like/view ratio (positive) and channel size (negative).

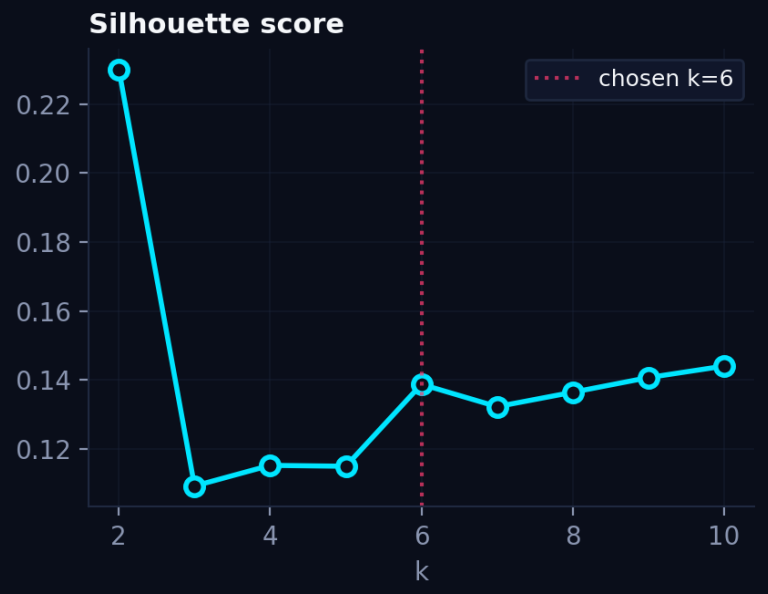
PC3-10 **Content shape**

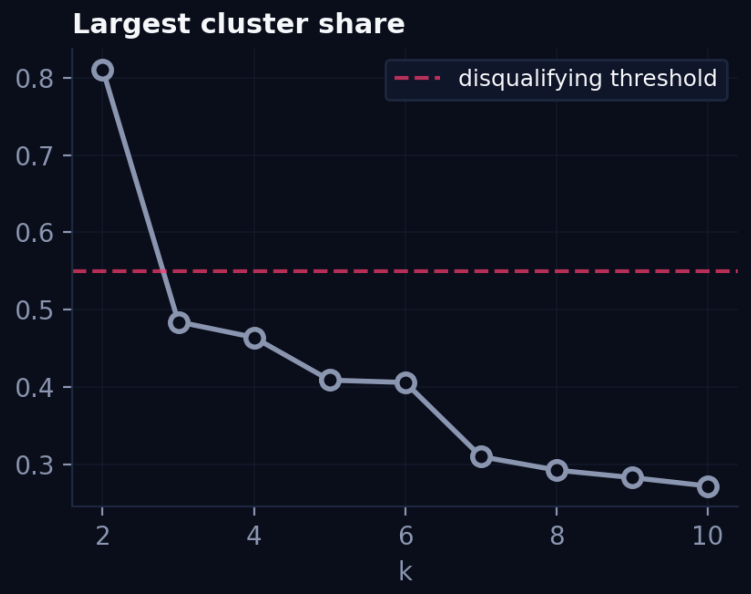
Duration, tag count, language, category — textural differences between archetypes.

Choosing k — three diagnostics, not silhouette alone

Silhouette would have picked k=2. We rejected that; here is why.







THE ELBOW

Inertia bends gently around k=4-6 with no sharp drop afterwards.

SILHOUETTE

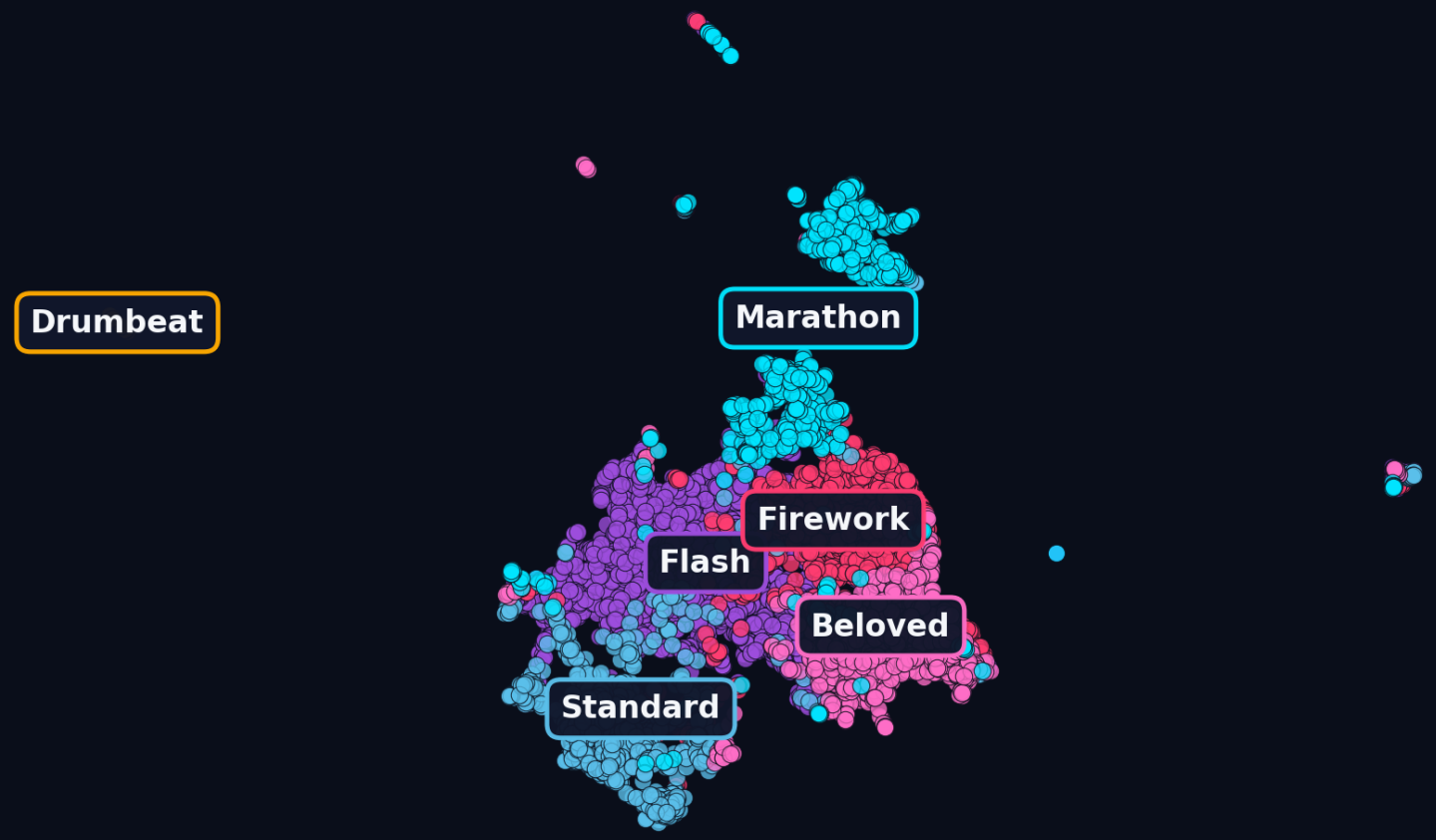
Highest at k=2, but degenerate (one cluster has 80% of videos). We rejected splits with > 55% imbalance.

CHOSEN: k=6

Best ARI between methods (0.55) at this k. Six clusters, four with clear archetype signatures.

Six archetypes emerge — four named, two fringe

K-Means at k=6 on 9,100 videos · methods agree (ARI = 0.55)



Firework n=1,598
Sustained big hits.
Music drops, Punjabi songs.

Beloved n=1,132
+2.1 like/view.
Tech creators, regional comedy.

Marathon n=1,119
+2.0 chart presence.
Tamil daily soaps stay forever.

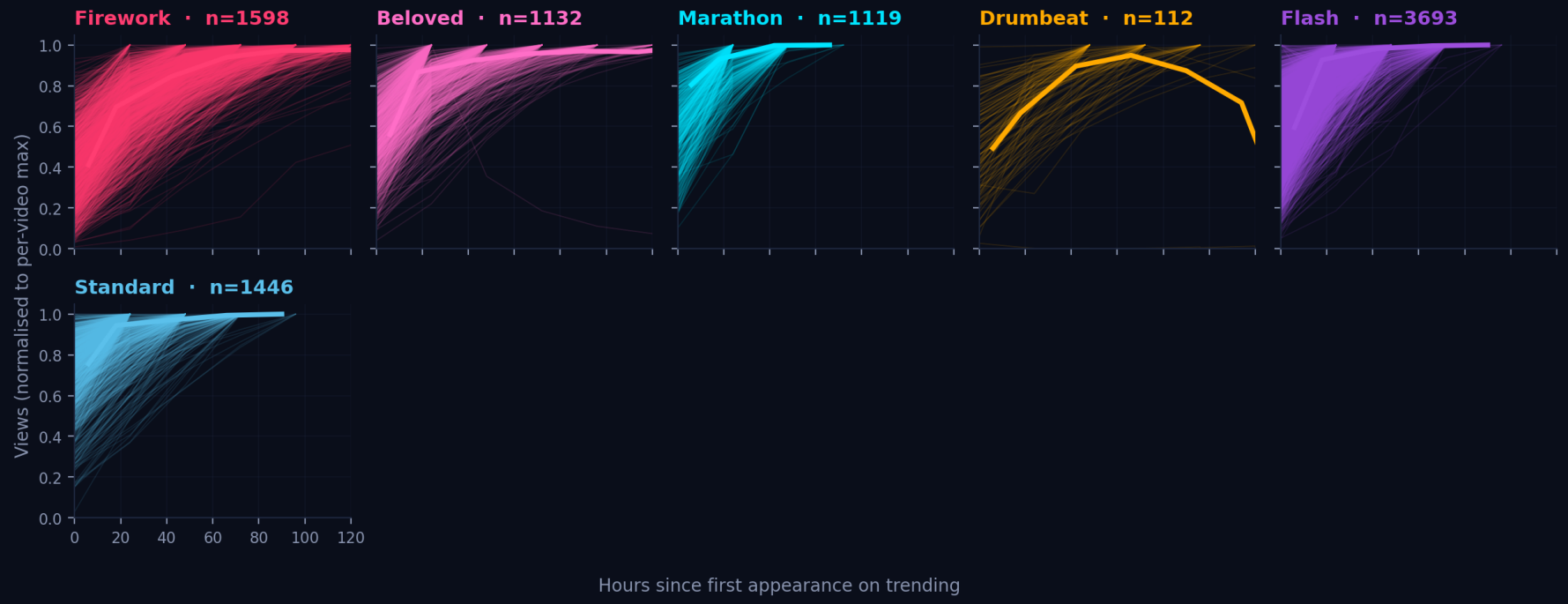
Drumbeat n=112
+6.5 returned-count.
News cycles re-enter the chart.

Flash n=3,693
Brief blips.
Most trending videos look like this.

Standard n=1,446
Baseline trender.
Daily television.

Each archetype has a distinct lifecycle shape

Average view-trajectories differ visibly. The archetypes are behavioural, not just statistical.

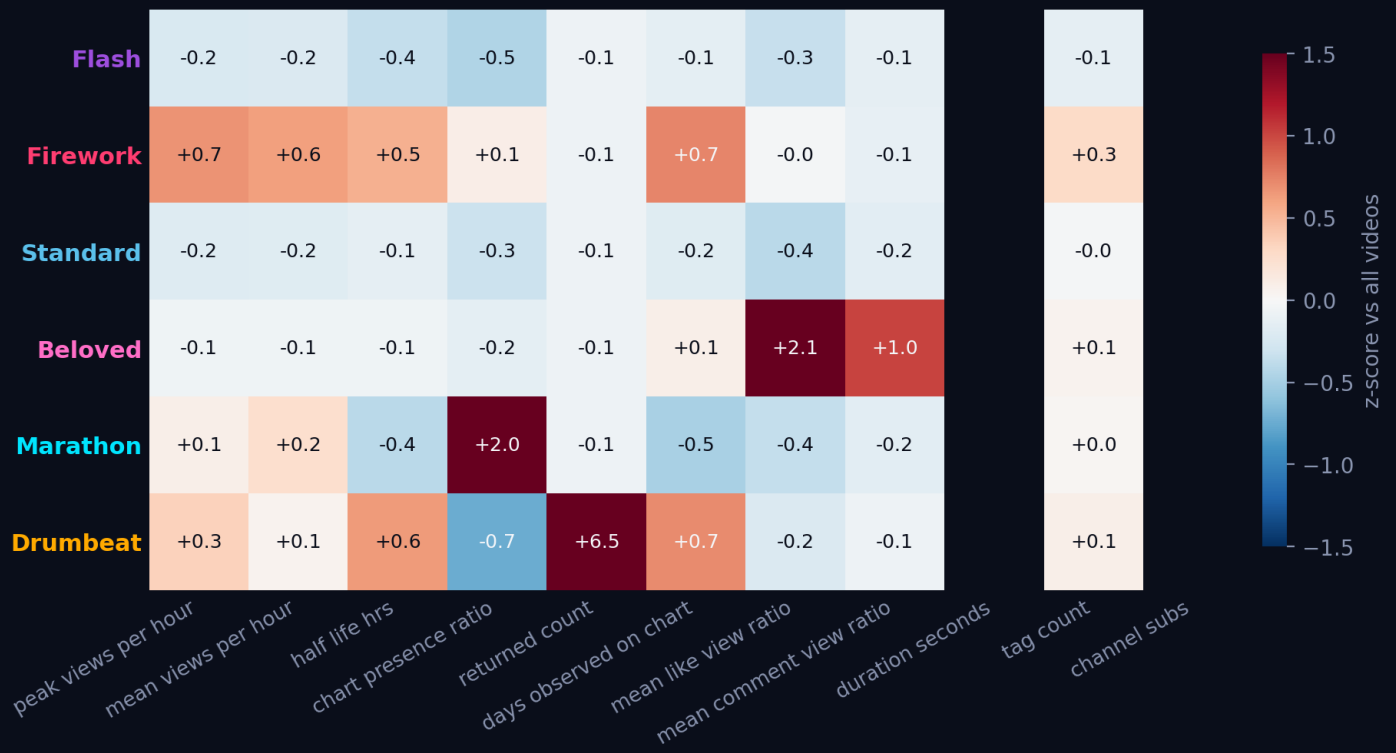


What you should notice

- Firework climbs in 24-48h to plateau, then slowly decays.
- Beloved videos look ordinary in views — engagement is what makes them special.
- Marathon plateaus near the top and stays there for days.
- Drumbeat is non-monotonic: dips and recoveries (chart re-entries).

What makes each archetype tick

Z-scored feature means per cluster. Reds = much more than average · blues = much less.



How to read this

- **Drumbeat:** +6.5 z-score on returned-count — videos that leave and re-enter the chart, the news-cycle pattern.
- **Marathon:** +2.0 chart-presence-ratio — daily soaps that stay on the chart for weeks.
- **Beloved:** +2.1 like/view ratio plus +1.0 comment/view — passionate audiences relative to size.
- **Firework:** Above-average velocity AND half-life — sustained big hits, not flash spikes.

What tags live in each archetype

Apriori on tag baskets · 1,015 rules at lift ≥ 3 · most distinctive tags per cluster shown below.

ARCHETYPE	DISTINCTIVE TAGS	TOP LIFT
Firework	punjabi romantic songs · speed records · latest punjabi songs 2018 · punjabi music	3.8×
Beloved	hyderabadi · balajimovies · technicalguruji · gaurav chaudhary	7.8×
Marathon	priyamanaval episode today · periyamanaval · priyamanaval today · priyamanaval episode	7.3×
Drumbeat	congress · telugu news · bjp · breaking news	4.7×
Standard	ozee · daily soap · watch online · anchor anasuya	2.8×
Flash	(no distinctive tags)	—

WHAT THIS TELLS US

Tags converge on the same archetypes the geometry found — Punjabi music IS Firework, Tamil daily soaps ARE Marathon, news cycles ARE Drumbeat. Two unsupervised methods arriving at the same partition is strong evidence the archetypes are real, not artefacts of one technique.

What we did — and what to take from it

1. Trending in India is not one phenomenon but at least four distinguishable archetypes — Marathon, Firework, Beloved, Drumbeat — plus a long tail of standard videos.
2. Lifecycle features (velocity, decay, retention, engagement) separate archetypes much more cleanly than static metadata alone.
3. K-Means and Agglomerative clustering agree (ARI = 0.55 on 9,100 videos) — the structure is real, not algorithm-dependent.
4. Association mining on tags surfaces the same archetypes (Punjabi songs↔Firework, daily soaps↔Marathon, news cycles↔Drumbeat) — convergent evidence.
5. Same pipeline runs on a fresh live API stream and on archival data without code changes — generalises to any region or window.

CAVEATS WE OWN

- Archival window is Nov 2017 - Jun 2018; trending behaviour today may differ.
- India only — archetype share will differ in other regions and YouTube eras.
- "Flash" cluster (n=3,693) is the residual catchall — interpret as "no distinctive lifecycle signature" rather than a true archetype.

WHERE THIS COULD GO

- Region comparison: India vs SG vs UK on the same pipeline.
- Time series: how does archetype share shift across weeks?
- Predictive layer: classify archetype from first-snapshot metadata only.

Thank you. Questions welcome.